Epoka University
Department of Computer Engineering

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Course Code: CEN 307

Project Title:

"Virtual Networking and Client-Server Architecture Application Deployment"

Group Number: 22

Submitted By:

- **Student 1 Name:** Sidrit Rrustemi
- **Student 2 Name:** Viola Zeneli
- **Student 3 Name:** Arsildo Veliu
- **Student 4 Name:** Megi Almadhi
- **Student 5 Name:** Artemisa Hasalami

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Instructor: Shkumbin Fida

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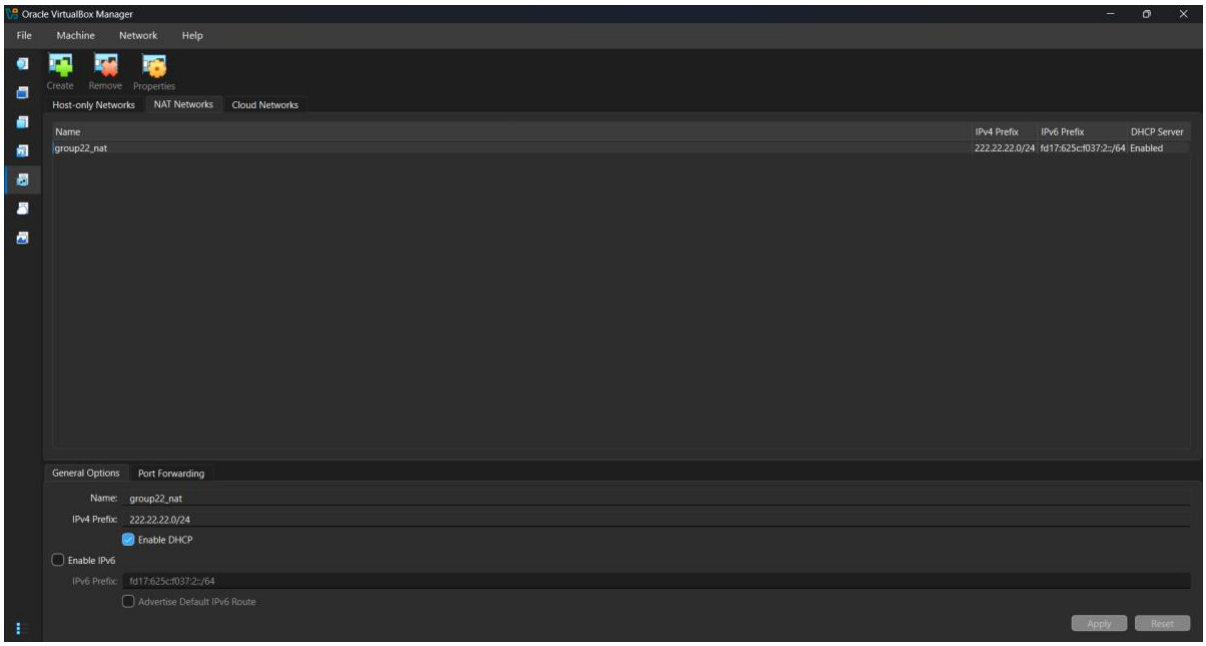
1

1. Deploy a Virtual Network NAT Network Using VirtualBox

A VirtualBox NAT Network was created to provide a private virtual LAN for all project virtual machines. The NAT Network is named **group22_nat** and is configured with the subnet **222.22.22.0/24**, where /24 corresponds to the subnet mask **255.255.255.0**. This configuration allows up to 254 usable host IP addresses within the same network segment. DHCP is enabled on the NAT Network, allowing virtual machines to automatically receive IP configuration parameters, including IP address, subnet mask, default gateway, and DNS server. The NAT Network is then started, activating the virtual router functionality that enables both internal communication between virtual machines and external connectivity to the Internet through Network Address Translation (NAT).

```
C:\Program Files\Oracle\VirtualBox>VBoxManage natnetwork add --netname group22_nat --network "222.22.22.0/24" --dhcp on

C:\Program Files\Oracle\VirtualBox>VBoxManage natnetwork start --netname group22_nat
```



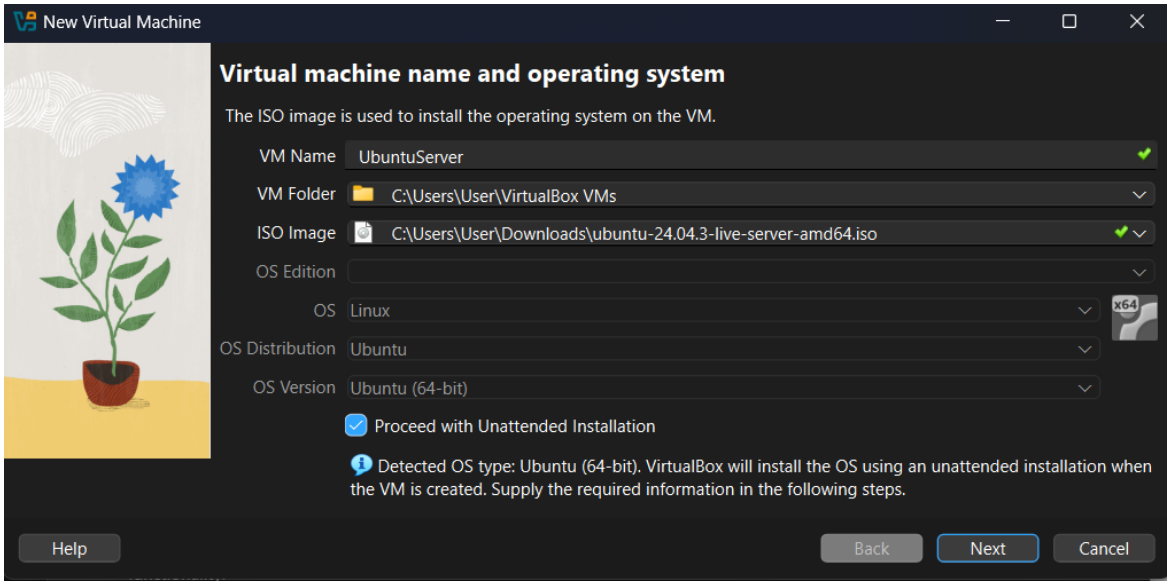
2

2. Install Guest OS Hosts on VirtualBox and Configure Networking

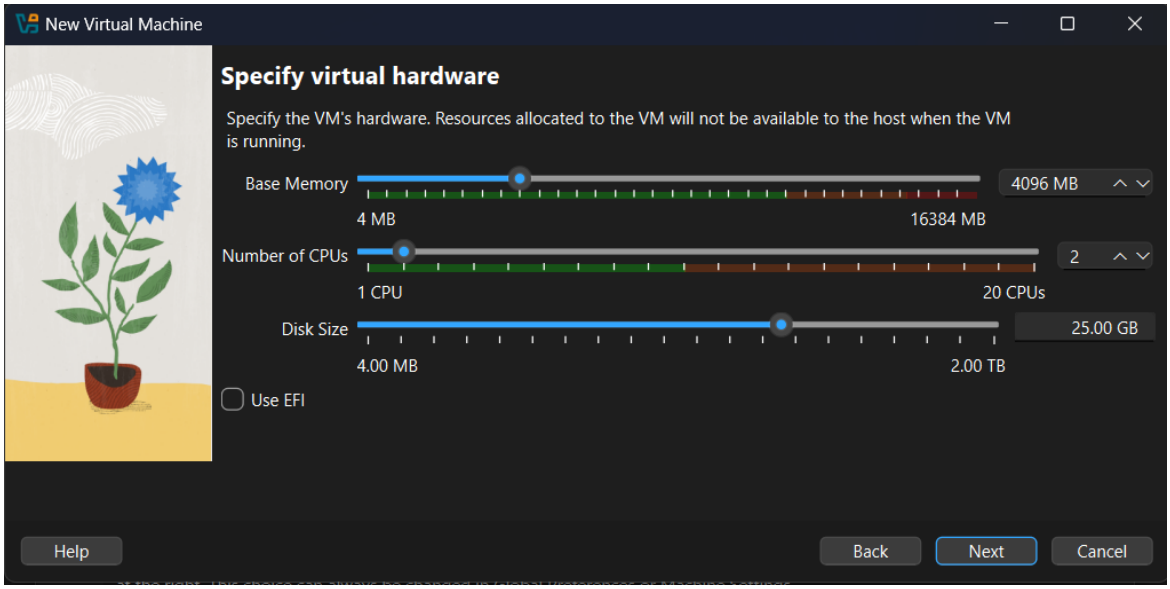
2.1 Ubuntu Server:

2.1.1 Install Ubuntu Server and set hostname group22-02 during installation

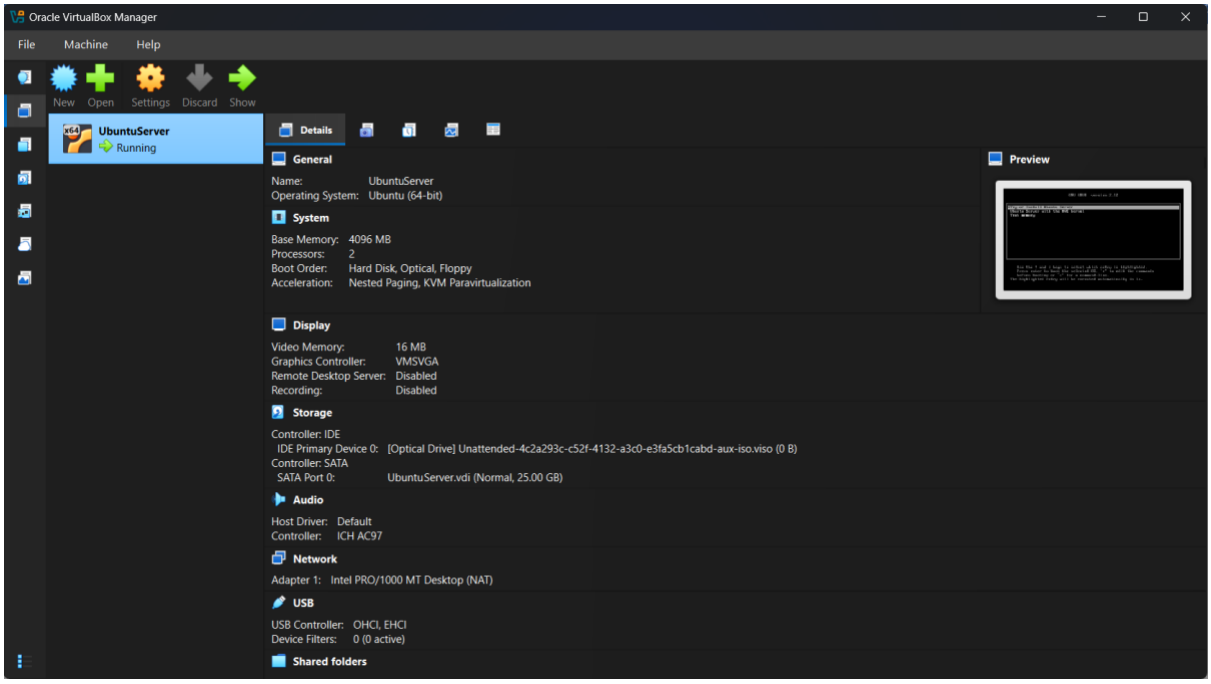
The Ubuntu Server virtual machine is created in VirtualBox using the Ubuntu Server installation image. Ubuntu Server is selected because it provides a lightweight, service-oriented operating system suitable for hosting network services such as web and SSH servers.



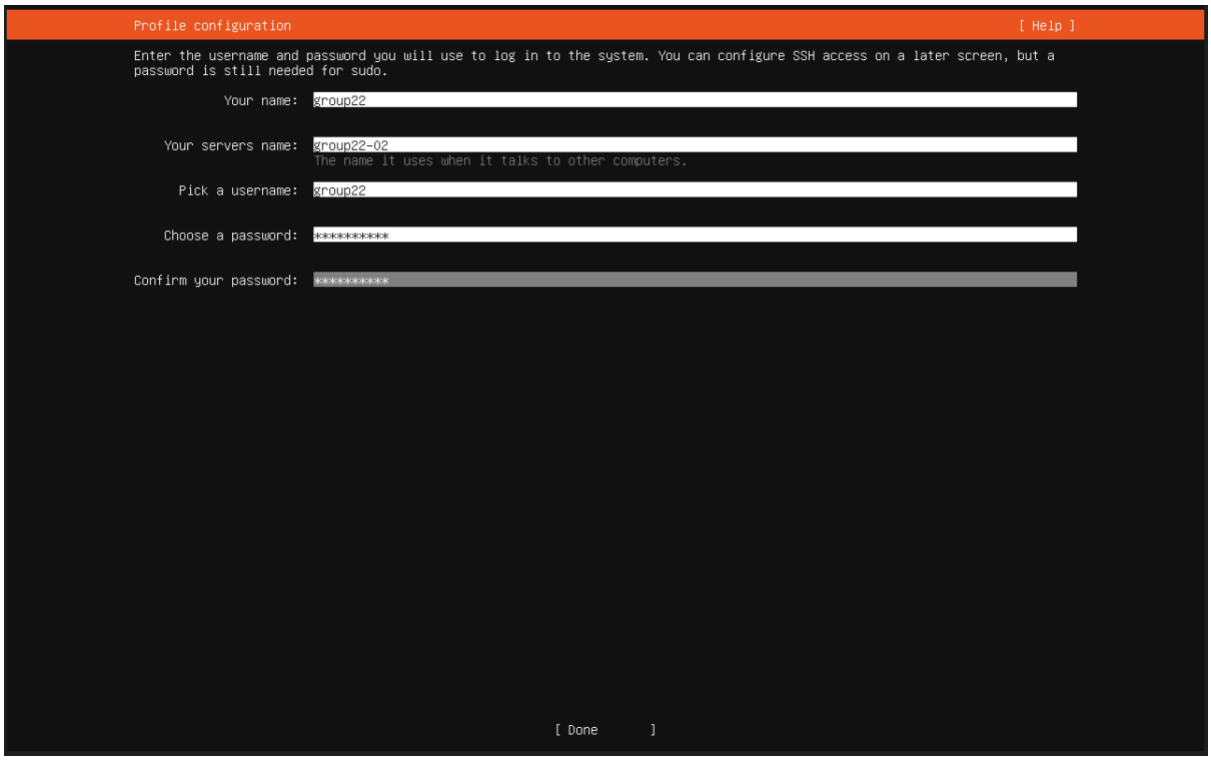
The virtual machine hardware is configured with **4096 MB of RAM**, **2 virtual CPUs**, and a **25 GB virtual disk**. These resources ensure adequate performance for running server-side network services without bottlenecks.



The Ubuntu Server installation process is initiated. This step installs the core operating system components required for server functionality, without a graphical interface, which reduces resource consumption and improves efficiency.



During installation, the system hostname is configured as **group22-02**, and a local user account named **group22** is created. The hostname uniquely identifies the server within the network and is later used for hostname-based communication and management.



After installation completion, the system boots successfully and displays the active network interface **enp0s3**, confirming that the server is operational and ready for network configuration and service deployment.

```
group22-02 login: group22
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-90-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Jan  7 10:57:13 AM UTC 2026

  System load:             0.23
  Usage of /:              42.1% of 11.21GB
  Memory usage:            6%
  Swap usage:              0%
  Processes:               119
  Users logged in:         0
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe76:b308

Expanded Security Maintenance for Applications is not enabled.

55 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status



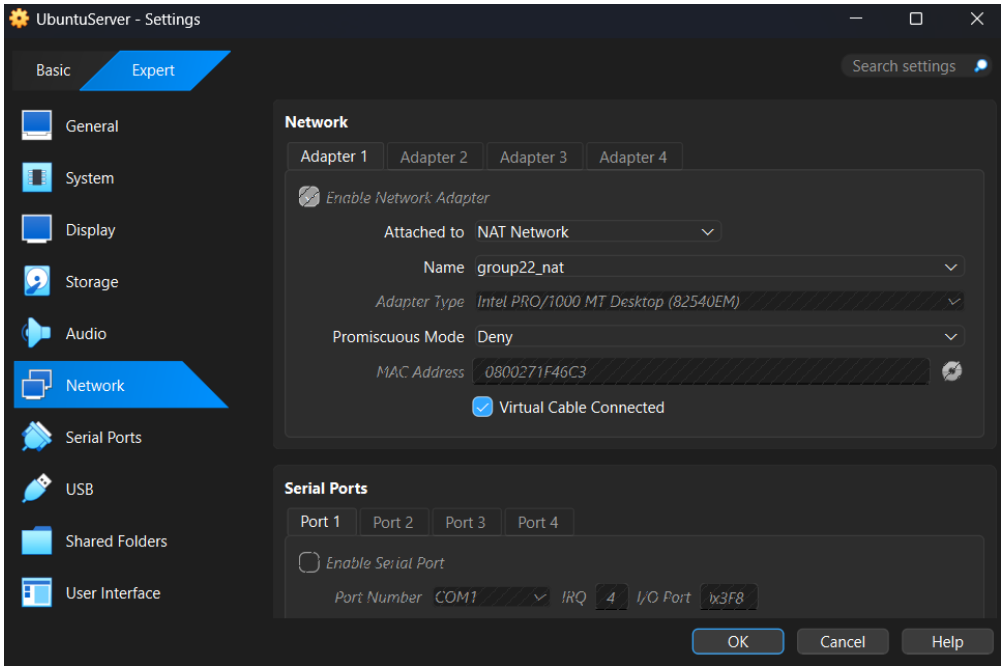
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```

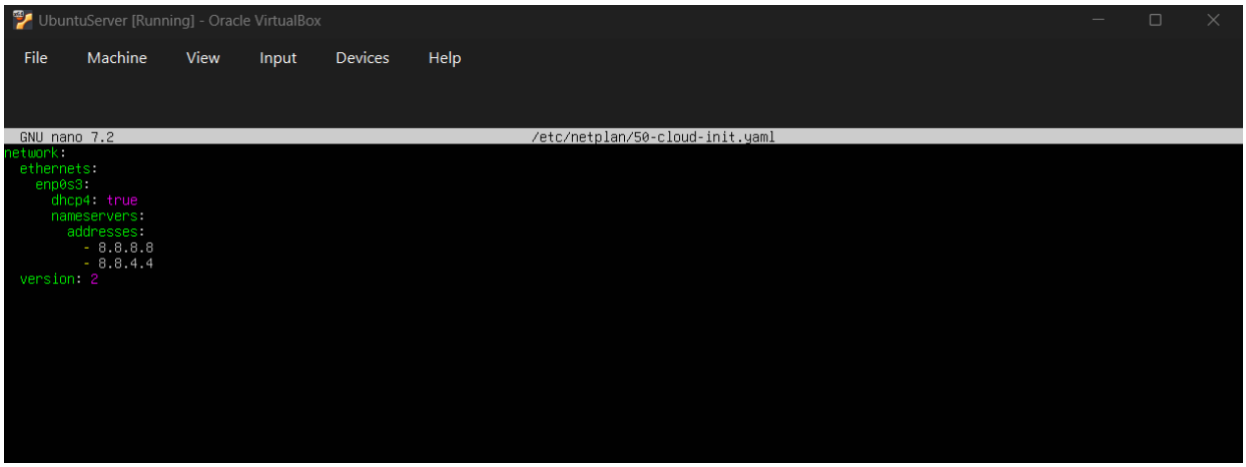
2.1.2 Configure the connected network interface to use DHCP

The Ubuntu Server network adapter is attached to the NAT Network **group22_nat**. This ensures that the server participates in the predefined subnet **222.22.22.0/24** and can communicate with other virtual machines connected to the same NAT Network. With DHCP enabled, the server automatically receives its network configuration.



2.1.3 Configure DNS on the network interface connected to NAT Network

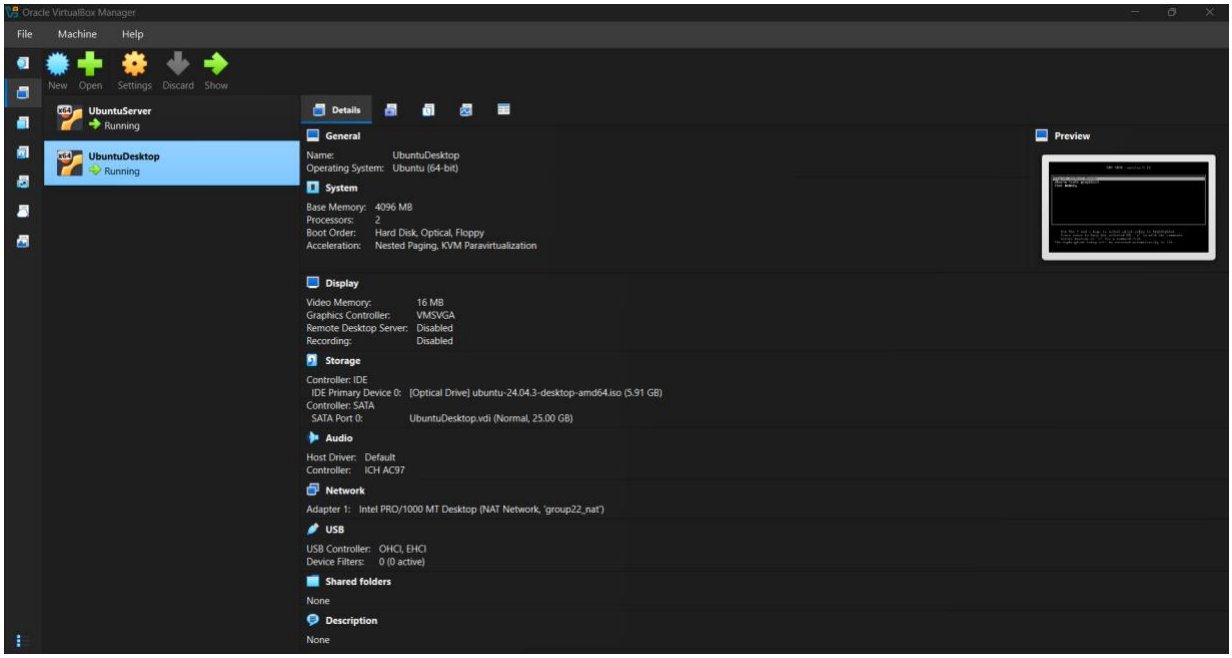
The network configuration file /etc/netplan/50-cloud-init.yaml specifies that the interface **enp0s3** uses DHCP for IPv4 addressing. DNS servers are explicitly configured as **8.8.8.8** and **8.8.4.4**, which are Google Public DNS servers. This configuration enables the server to resolve domain names into IP addresses, which is required for accessing external hosts and services using domain names.



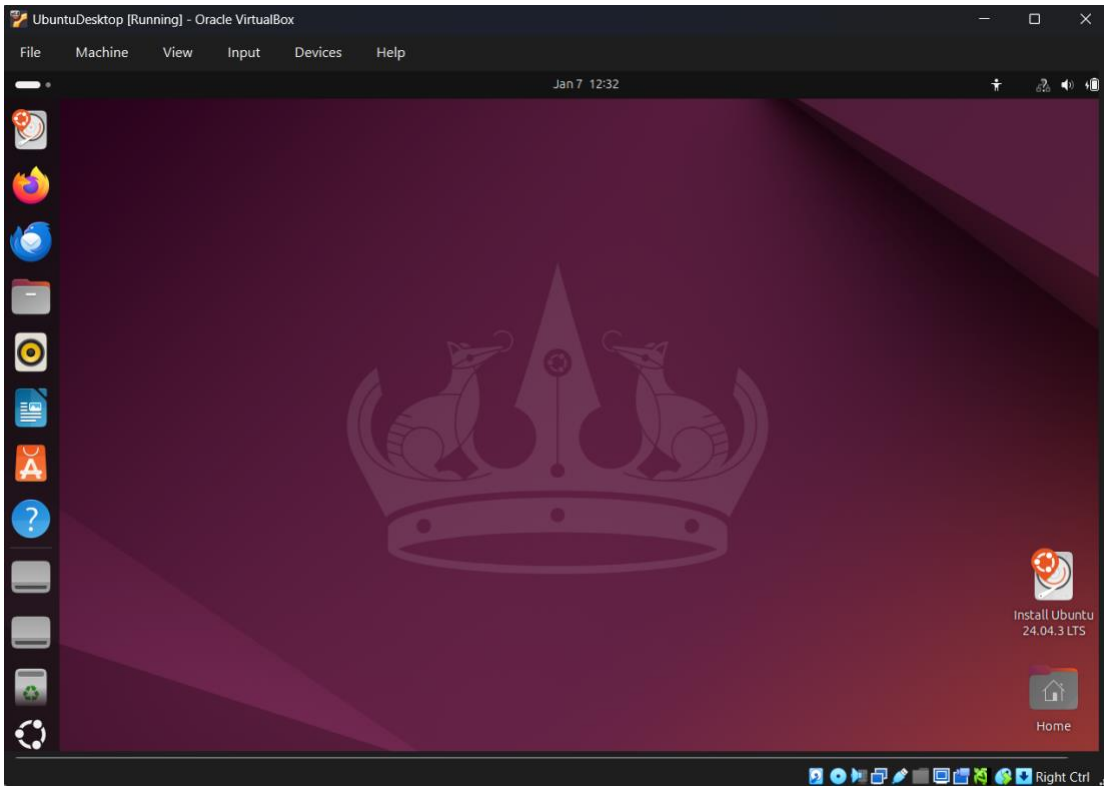
2.2 Ubuntu Desktop:

2.2.1 Install Ubuntu Desktop and set hostname group22-01 during installation

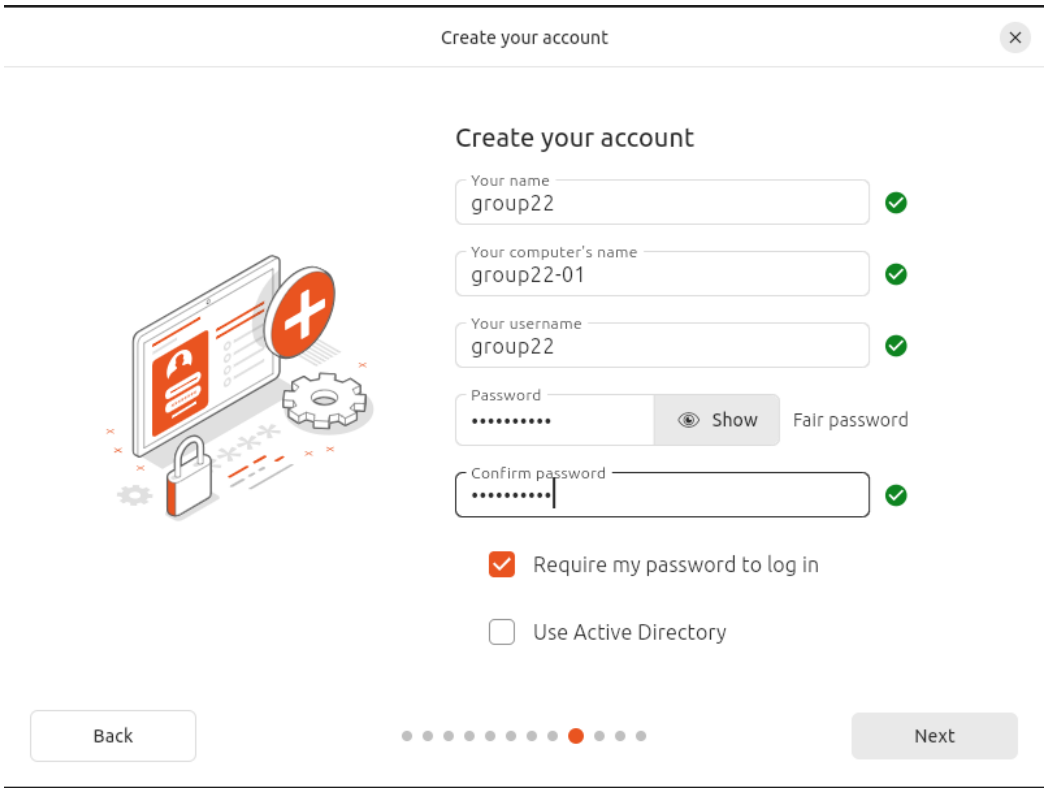
The Ubuntu Desktop virtual machine is created using the Ubuntu Desktop installation image and connected to the NAT Network **group22_nat**. Ubuntu Desktop provides a graphical user interface required for client-side tools such as web browsers and packet analysis software.



The Ubuntu Desktop installation environment is loaded, allowing system installation with a graphical interface suitable for end-user interaction.

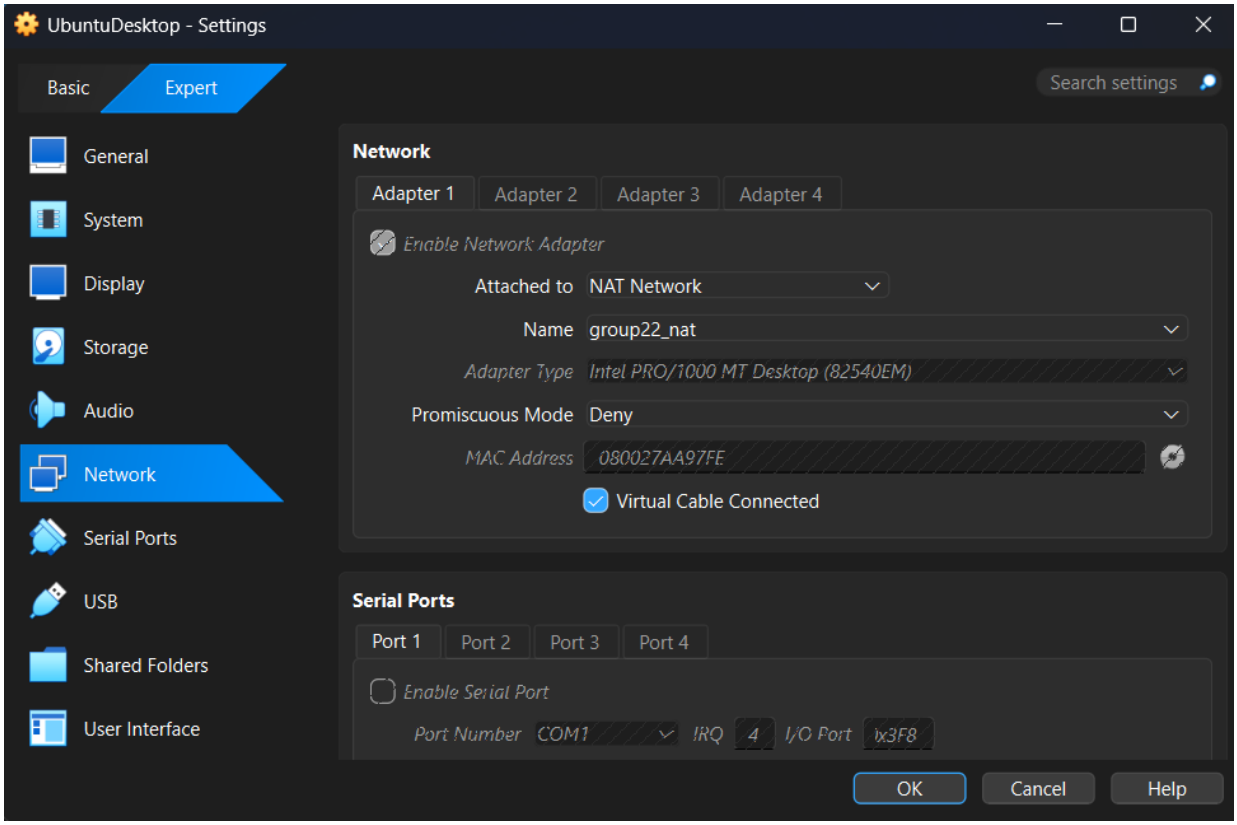


During installation, the hostname is set to **group22-01**, and the user account **group22** is created. The hostname distinguishes the client machine from the server within the same network.



2.2.2 Configure DNS on the network interface connected to NAT Network

The network adapter is confirmed to be attached to the NAT Network **group22_nat**, ensuring the desktop system is placed in the same subnet as the server for direct communication.



3

3. Install Client Server Applications

3.1 On Ubuntu Server (group22-02):

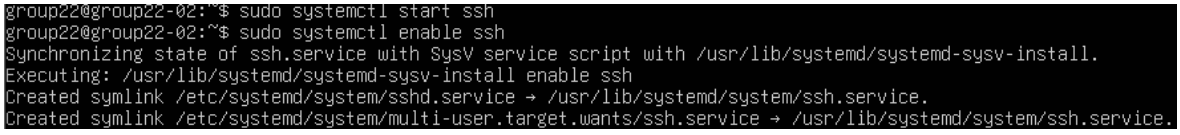
3.1.1 Install Nginx Web Server on Apache Web Server

The Nginx web server service is installed and verified as active and running. Nginx functions as the **HTTP web server**, listening for incoming HTTP requests on TCP port **80** and responding with web content. This server represents the application-layer service provider in the client–server architecture implemented in this project.



3.1.2 Ensure OpenSSH Server is installed

The OpenSSH server service is started and enabled to run automatically at system boot. OpenSSH provides secure remote access to the server using the SSH protocol, which operates over TCP port **22** and ensures encrypted communication between client and server.



3.2 On Ubuntu Desktop (group22-01):

3.2.1 Install Google Chrome or Mozilla Firefox

The Firefox web browser is installed and verified as the latest available version. The browser functions as the **HTTP client**, sending HTTP requests to the web server and receiving HTTP responses that are rendered for the user.

```
group22@group22-01:~$ sudo apt install firefox -y
[sudo] password for group22:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
firefox is already the newest version (1:1snap1-0ubuntu5).
firefox set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 148 not upgraded.
```

3.2.2 Install the OpenSSH Client

The SSH client utility is available on the Ubuntu Desktop system. This client initiates SSH connections to remote servers, enabling secure command-line access to the Ubuntu Server.

```
group22@group22-01:~$ ssh
usage: ssh [-46AaCfGgKkMNnqsTtVvXxYy] [-B bind_interface] [-b bind_address]
           [-c cipher_spec] [-D [bind_address:]port] [-E log_file]
           [-e escape_char] [-F configfile] [-I pkcs11] [-i identity_file]
           [-J destination] [-L address] [-l login_name] [-m mac_spec]
           [-O ctl_cmd] [-o option] [-P tag] [-p port] [-R address]
           [-S ctl_path] [-W host:port] [-w local_tun[:remote_tun]]
           destination [command [argument ...]]
       ssh [-Q query_option]
```

4

4. Test Network Connectivity Between Hosts

4.1 Verify IP addresses on both Ubuntu Desktop and Server using **ip a** or **ifconfig**

The network interface **enp0s3** on the Ubuntu Desktop is assigned the IP address **222.22.22.6/24** via DHCP. The broadcast address is **222.22.22.255**, confirming the system is correctly configured within the subnet **222.22.22.0/24**.

```
group22@group22-01:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:aa:97:fe brd ff:ff:ff:ff:ff:ff
    inet 222.22.22.6/24 brd 222.22.22.255 scope global dynamic noprefixroute enp
0s3
       valid_lft 526sec preferred_lft 526sec
    inet6 fe80::a00:27ff:feaa:97fe/64 scope link
       valid_lft forever preferred_lft forever
```

The network interface **enp0s3** on the Ubuntu Server is assigned the IP address **222.22.22.5/24**, also via DHCP. This confirms both hosts reside in the same Layer-3 network and can communicate directly.

```
group22@group22-02:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:76:b3:08 brd ff:ff:ff:ff:ff:ff
    inet 222.22.22.5/24 metric 100 brd 222.22.22.255 scope global dynamic enp0s3
       valid_lft 407sec preferred_lft 407sec
    inet6 fe80::a00:27ff:fe76:b308/64 scope link
       valid_lft forever preferred_lft forever
```

4.2 Test local connectivity:

4.2.1 Ping group22-02 host from group22-01 host

ICMP Echo Requests are sent from **group22-01** to the server IP **222.22.22.5**. All packets are successfully received with **0% packet loss**, and round-trip time values are recorded. This confirms successful Layer-3 connectivity between client and server.

```
group22@group22-01:~$ ping 222.22.22.5
PING 222.22.22.5 (222.22.22.5) 56(84) bytes of data.
64 bytes from 222.22.22.5: icmp_seq=1 ttl=64 time=1.76 ms
64 bytes from 222.22.22.5: icmp_seq=2 ttl=64 time=1.33 ms
64 bytes from 222.22.22.5: icmp_seq=3 ttl=64 time=0.912 ms
64 bytes from 222.22.22.5: icmp_seq=4 ttl=64 time=1.13 ms
64 bytes from 222.22.22.5: icmp_seq=5 ttl=64 time=1.04 ms
64 bytes from 222.22.22.5: icmp_seq=6 ttl=64 time=1.12 ms
64 bytes from 222.22.22.5: icmp_seq=7 ttl=64 time=0.660 ms
64 bytes from 222.22.22.5: icmp_seq=8 ttl=64 time=1.11 ms
64 bytes from 222.22.22.5: icmp_seq=9 ttl=64 time=1.07 ms
64 bytes from 222.22.22.5: icmp_seq=10 ttl=64 time=0.978 ms
64 bytes from 222.22.22.5: icmp_seq=11 ttl=64 time=1.36 ms
64 bytes from 222.22.22.5: icmp_seq=12 ttl=64 time=1.34 ms
64 bytes from 222.22.22.5: icmp_seq=13 ttl=64 time=2.01 ms
64 bytes from 222.22.22.5: icmp_seq=14 ttl=64 time=1.99 ms
64 bytes from 222.22.22.5: icmp_seq=15 ttl=64 time=1.34 ms
64 bytes from 222.22.22.5: icmp_seq=16 ttl=64 time=1.03 ms
64 bytes from 222.22.22.5: icmp_seq=17 ttl=64 time=0.566 ms
64 bytes from 222.22.22.5: icmp_seq=18 ttl=64 time=1.38 ms
64 bytes from 222.22.22.5: icmp_seq=19 ttl=64 time=1.01 ms
64 bytes from 222.22.22.5: icmp_seq=20 ttl=64 time=1.13 ms
64 bytes from 222.22.22.5: icmp_seq=21 ttl=64 time=0.778 ms
^C
--- 222.22.22.5 ping statistics ---
21 packets transmitted, 21 received, 0% packet loss, time 20060ms
rtt min/avg/max/mdev = 0.566/1.191/2.010/0.368 ms
```

4.2.2 Ping group22-01 host from group22-02 host

ICMP Echo Requests are sent from **group22-02** to the desktop IP **222.22.22.6**. Successful replies confirm bidirectional communication between both hosts.

```
group22@group22-02:~$ ping 222.22.22.6
PING 222.22.22.6 (222.22.22.6) 56(84) bytes of data.
64 bytes from 222.22.22.6: icmp_seq=1 ttl=64 time=1.26 ms
64 bytes from 222.22.22.6: icmp_seq=2 ttl=64 time=2.34 ms
64 bytes from 222.22.22.6: icmp_seq=3 ttl=64 time=1.37 ms
64 bytes from 222.22.22.6: icmp_seq=4 ttl=64 time=1.72 ms
64 bytes from 222.22.22.6: icmp_seq=5 ttl=64 time=2.02 ms
64 bytes from 222.22.22.6: icmp_seq=6 ttl=64 time=2.88 ms
64 bytes from 222.22.22.6: icmp_seq=7 ttl=64 time=1.08 ms
64 bytes from 222.22.22.6: icmp_seq=8 ttl=64 time=2.64 ms
64 bytes from 222.22.22.6: icmp_seq=9 ttl=64 time=2.93 ms
64 bytes from 222.22.22.6: icmp_seq=10 ttl=64 time=1.48 ms
64 bytes from 222.22.22.6: icmp_seq=11 ttl=64 time=3.88 ms
```

4.3 Test external connectivity using public hosts/web site using ping and traceroute

4.3.1 Ping a public host from group22-01 host

The desktop system successfully resolves **www.google.com** to an external IP address and receives ICMP Echo Replies with **0% packet loss**. This confirms correct DNS resolution and outbound Internet connectivity through the NAT Network.

```
group22@group22-01:~$ ping www.google.com
PING www.google.com (74.125.138.147) 56(84) bytes of data.
64 bytes from yi-in-f147.1e100.net (74.125.138.147): icmp_seq=1 ttl=255 time=142
 ms
64 bytes from yi-in-f147.1e100.net (74.125.138.147): icmp_seq=2 ttl=255 time=141
 ms
64 bytes from yi-in-f147.1e100.net (74.125.138.147): icmp_seq=3 ttl=255 time=141
 ms
64 bytes from yi-in-f147.1e100.net (74.125.138.147): icmp_seq=4 ttl=255 time=142
 ms
64 bytes from yi-in-f147.1e100.net (74.125.138.147): icmp_seq=5 ttl=255 time=141
 ms
64 bytes from yi-in-f147.1e100.net (74.125.138.147): icmp_seq=6 ttl=255 time=140
 ms
64 bytes from yi-in-f147.1e100.net (74.125.138.147): icmp_seq=7 ttl=255 time=142
 ms
64 bytes from yi-in-f147.1e100.net (74.125.138.147): icmp_seq=8 ttl=255 time=141
 ms
64 bytes from yi-in-f147.1e100.net (74.125.138.147): icmp_seq=9 ttl=255 time=141
 ms
64 bytes from yi-in-f147.1e100.net (74.125.138.147): icmp_seq=10 ttl=255 time=14
2 ms
64 bytes from yi-in-f147.1e100.net (74.125.138.147): icmp_seq=11 ttl=255 time=14
0 ms
64 bytes from yi-in-f147.1e100.net (74.125.138.147): icmp_seq=12 ttl=255 time=14
1 ms
^C
--- www.google.com ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11016ms
rtt min/avg/max/mdev = 140.385/141.248/142.469/0.655 ms
```

4.3.2 Run traceroute to a public host/web site from both hosts

Ubuntu Desktop

Traceroute shows the first hop as the NAT Network gateway **222.22.22.1**, which represents the virtual router. Subsequent hops are not displayed due to ICMP filtering, which is common on public networks.

```
group22@group22-01:~$ traceroute www.google.com
traceroute to www.google.com (74.125.138.106), 30 hops max, 60 byte packets
 1  _gateway (222.22.22.1)  0.935 ms  0.829 ms  0.772 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

Ubuntu Server

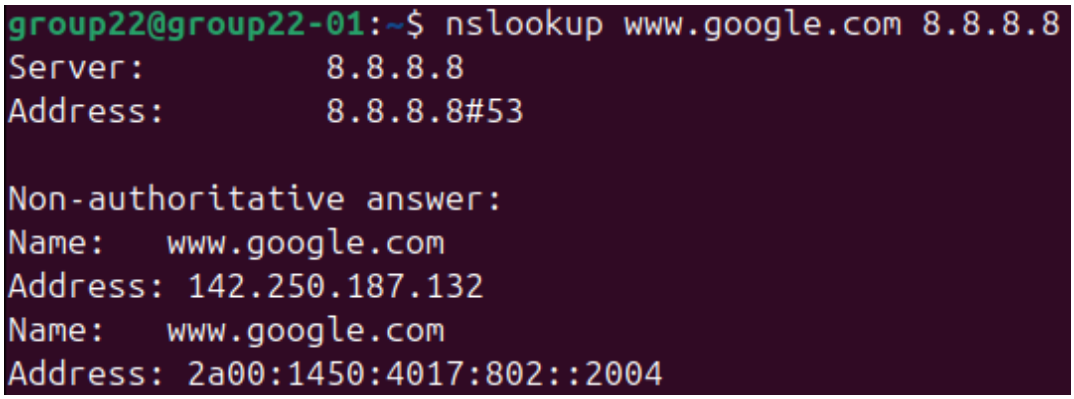
Traceroute from the server produces similar results, confirming both systems use the same default gateway for external routing.



4.4 Perform DNS resolution using nslookup using public DNS server on your chosen hostname or web site

Ubuntu Desktop

A DNS query is sent to **8.8.8.8**, and valid IPv4 and IPv6 addresses for **www.google.com** are returned. This confirms correct DNS operation.



Ubuntu Server

The server performs the same DNS query and receives valid responses, demonstrating functional DNS resolution on both hosts.

```
group22@group22-02:~$ nslookup www.google.com 8.8.8.8
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   www.google.com
Address: 216.58.204.228
Name:   www.google.com
Address: 2a00:1450:4002:415::2004
```

4.5 Use the OpenSSH client to connect remotely from Ubuntu Desktop to Ubuntu Server

An SSH connection is established from **group22-01** to the server at **222.22.22.5**. Authentication succeeds, confirming secure remote access functionality.

```
group22@group22-01:~$ ssh group22@222.22.22.5
The authenticity of host '222.22.22.5 (222.22.22.5)' can't be established.
ED25519 key fingerprint is SHA256:ffBbcDfMjUUuJuT+pncbQ3OF8hJuDzBI/uGHScekM2I.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '222.22.22.5' (ED25519) to the list of known hosts.
group22@222.22.22.5's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-90-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sun Jan 18 09:06:38 PM UTC 2026

  System load:  0.0               Processes:               122
  Usage of /:   48.9% of 11.21GB  Users logged in:         1
  Memory usage: 5%                IPv4 address for enp0s3: 222.22.22.5
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

55 updates can be applied immediately.
2 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

8 additional security updates can be applied with ESM Apps.
```

The login information confirms successful remote access and records the client IP address, demonstrating proper SSH session establishment.

```
Learn more about enabling ESM Apps service at https://ubuntu.com/esm


Last login: Wed Jan 14 10:27:21 2026 from 222.22.22.6
```

4.6 Assign static IP addresses to both Ubuntu Desktop and Ubuntu Server and repeat steps 4.2 and 4.3

The Ubuntu Server network configuration is modified to disable DHCP and assign the static IP **222.22.22.10/24**, with default gateway **222.22.22.1** and DNS servers **8.8.8.8** and **8.8.4.4**.

```
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 222.22.22.10/24
      gateway4: 222.22.22.1
      nameservers:
        addresses:
          - 8.8.8.8
          - 8.8.4.4
```

Verification confirms the server interface is assigned the static IP **222.22.22.10/24**.

```
group22@group22-02:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:76:b3:08 brd ff:ff:ff:ff:ff:ff
    inet 222.22.22.10/24 brd 222.22.22.255 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe76:b308/64 scope link
       valid_lft forever preferred_lft forever
```

The Ubuntu Desktop network configuration is updated to use the static IP **222.22.22.11/24**, with the same gateway and DNS servers.

```
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 222.22.22.11/24
      gateway4: 222.22.22.1
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
```

Verification confirms the desktop interface is assigned the static IP **222.22.22.11/24**.

```
group22@group22-01:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:aa:97:fe brd ff:ff:ff:ff:ff:ff
    inet 222.22.22.11/24 brd 222.22.22.255 scope global noprefixroute enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feaa:97fe/64 scope link
       valid_lft forever preferred_lft forever
```

The file /etc/hosts on the Ubuntu Desktop maps **222.22.22.10** to the hostname **group22-02**, enabling local hostname resolution without DNS.

```
127.0.0.1 localhost
127.0.1.1 group22-01

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

222.22.22.10 group22-02
```

The /etc/hosts file on the Ubuntu Server maps **222.22.22.11** to **group22-01**, enabling mutual hostname resolution.

```
127.0.0.1 localhost
127.0.1.1 group22-02

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

222.22.22.11 group22-01
```

Ping tests using both IP addresses and hostnames succeed in both directions, confirming correct static addressing and hostname resolution.

```
group22@group22-01:~$ ping -c 4 222.22.22.10
PING 222.22.22.10 (222.22.22.10) 56(84) bytes of data.
64 bytes from 222.22.22.10: icmp_seq=1 ttl=64 time=4.70 ms
64 bytes from 222.22.22.10: icmp_seq=2 ttl=64 time=1.90 ms
64 bytes from 222.22.22.10: icmp_seq=3 ttl=64 time=2.00 ms
64 bytes from 222.22.22.10: icmp_seq=4 ttl=64 time=2.21 ms

--- 222.22.22.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.899/2.701/4.699/1.158 ms
```

```
group22@group22-01:~$ ping -c 4 group22-02
PING group22-02 (222.22.22.10) 56(84) bytes of data.
64 bytes from group22-02 (222.22.22.10): icmp_seq=1 ttl=64 time=1.82 ms
64 bytes from group22-02 (222.22.22.10): icmp_seq=2 ttl=64 time=2.62 ms
64 bytes from group22-02 (222.22.22.10): icmp_seq=3 ttl=64 time=2.30 ms
64 bytes from group22-02 (222.22.22.10): icmp_seq=4 ttl=64 time=2.60 ms

--- group22-02 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 1.817/2.332/2.616/0.323 ms
```

```
group22@group22-02:~$ ping -c 4 222.22.22.11
PING 222.22.22.11 (222.22.22.11) 56(84) bytes of data.
64 bytes from 222.22.22.11: icmp_seq=1 ttl=64 time=3.17 ms
64 bytes from 222.22.22.11: icmp_seq=2 ttl=64 time=3.18 ms
64 bytes from 222.22.22.11: icmp_seq=3 ttl=64 time=2.14 ms
64 bytes from 222.22.22.11: icmp_seq=4 ttl=64 time=1.84 ms

--- 222.22.22.11 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3634ms
rtt min/avg/max/mdev = 1.842/2.582/3.178/0.602 ms
```

```
group22@group22-02:~$ ping -c 4 group22-01
PING group22-01 (222.22.22.11) 56(84) bytes of data.
64 bytes from group22-01 (222.22.22.11): icmp_seq=1 ttl=64 time=2.07 ms
64 bytes from group22-01 (222.22.22.11): icmp_seq=2 ttl=64 time=3.40 ms
64 bytes from group22-01 (222.22.22.11): icmp_seq=3 ttl=64 time=3.48 ms
64 bytes from group22-01 (222.22.22.11): icmp_seq=4 ttl=64 time=4.25 ms

--- group22-01 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3017ms
rtt min/avg/max/mdev = 2.069/3.297/4.247/0.783 ms
```

Ping and traceroute tests to public hosts remain successful, demonstrating that static IP configuration does not affect external connectivity.

```
group22@group22-01:~$ ping -c 4 www.google.com
PING www.google.com (142.250.187.132) 56(84) bytes of data.
64 bytes from sof02s45-in-f4.1e100.net (142.250.187.132): icmp_seq=1 ttl=255 time=21.3 ms
64 bytes from sof02s45-in-f4.1e100.net (142.250.187.132): icmp_seq=2 ttl=255 time=22.8 ms
64 bytes from sof02s45-in-f4.1e100.net (142.250.187.132): icmp_seq=3 ttl=255 time=21.5 ms
64 bytes from sof02s45-in-f4.1e100.net (142.250.187.132): icmp_seq=4 ttl=255 time=20.5 ms

--- www.google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 20.467/21.519/22.802/0.835 ms
```

```
group22@group22-01:~$ traceroute www.google.com
traceroute to www.google.com (142.250.187.132), 30 hops max, 60 byte packets
 1  _gateway (222.22.22.1)  4.945 ms  4.842 ms  4.538 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

```
group22@group22-02:~$ traceroute www.google.com
traceroute to www.google.com (216.58.204.132), 30 hops max, 60 byte packets
 1  _gateway (222.22.22.1)  0.786 ms  1.233 ms  0.879 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

The Nginx service remains active after static IP configuration, confirming service stability.

```
group22@group22-02:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-01-14 09:07:53 UTC; 1h 9min ago
       Docs: man:nginx(8)
   Main PID: 873 (nginx)
      Tasks: 3 (limit: 4605)
     Memory: 3.7M (peak: 3.7M)
        CPU: 34ms
     CGroup: /system.slice/nginx.service
             ├─873 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─874 "nginx: worker process"
             └─875 "nginx: worker process"

Jan 14 09:07:53 group22-02 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy serv
Jan 14 09:07:53 group22-02 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy serve
```

A local HTTP request using curl verifies that the web server is functioning correctly on the server system.

```
group22@group22-02:~$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

4.7 Map Ubuntu Server hostname with IP (simulate local DNS) on for Ubuntu Desktop on /etc/hosts configuration file

The **/etc/hosts** file on the Ubuntu Desktop was configured to map the Ubuntu Server's IP address (222.22.22.10) to its hostname (group22-02). This simulates a local DNS resolution, allowing the server to be reached by hostname instead of IP within the virtual network.

```
group22@group22-01: ~
  GNU nano 7.2                          /etc/hosts
127.0.0.1 localhost
127.0.1.1 group22-01

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

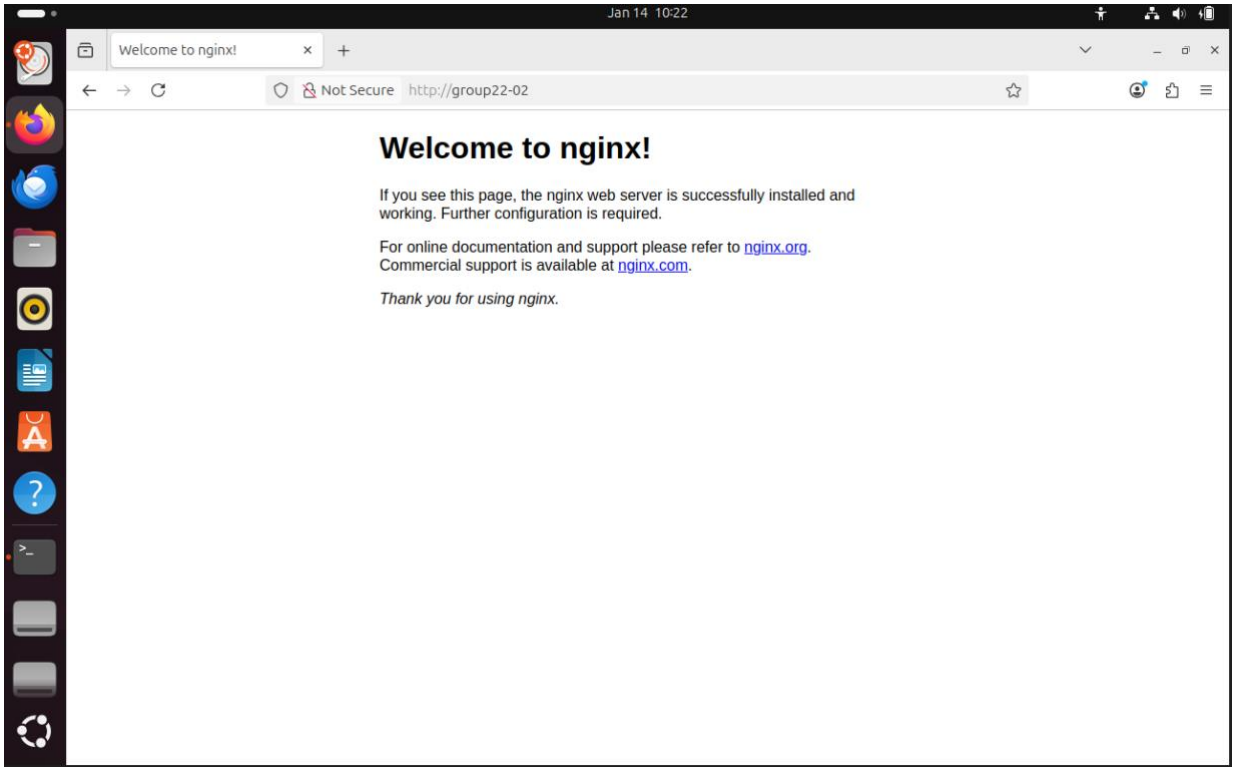
222.22.22.10 group22-02
```

5

5. Test Client Server Applications

5.1 From Ubuntu Desktop, access the default Nginx/Apache web page on Ubuntu Server via IP or hostname

The Ubuntu Desktop browser accesses the web server using the hostname **group22-02** and successfully displays the default Nginx page. Nginx functions as the **web server**, handling HTTP requests from the Ubuntu Desktop browser, while the browser operates as the **HTTP client**, receiving and rendering the server's response.



5.2 Use the OpenSSH client on Ubuntu Desktop to connect to the OpenSSH server on Ubuntu Server via IP or hostname

An SSH connection is successfully established using the static IP **222.22.22.10**, confirming that secure remote access works consistently with static addressing.

```
group22@group22-01:~$ ssh group22@222.22.22.10
group22@222.22.22.10's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-90-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Jan 29 09:34:46 PM UTC 2026

  System load:  0.07               Processes:               125
  Usage of /:   49.2% of 11.21GB   Users logged in:         1
  Memory usage: 5%                 IPv4 address for enp0s3: 222.22.22.10
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

55 updates can be applied immediately.
2 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

8 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
```

```
Last login: Sun Jan 18 21:52:08 2026 from 222.22.22.11
```

6

6. Install Networking Tools

6.1 On Ubuntu Server install:

6.1.1 net-tools

The net-tools package is installed, providing classic networking utilities such as ifconfig and netstat for interface and connection inspection.

```
group22@group22-02:~$ sudo apt update
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://al.archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://al.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://al.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,404 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [920 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [209 kB]
Fetched 2,659 kB in 2s (1,650 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
57 packages can be upgraded. Run 'apt list --upgradable' to see them.
group22@group22-02:~$ sudo apt install net-tools -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
net-tools is already the newest version (2.10-0.1ubuntu4.4).
0 upgraded, 0 newly installed, 0 to remove and 57 not upgraded.
```

6.1.2 iperf3

The iperf3 service is enabled and running in server mode. This allows the system to accept throughput test connections from a client.

```
group22@group22-02:~$ sudo systemctl status iperf3
● iperf3.service - iperf3 server
     Loaded: loaded (/usr/lib/systemd/system/iperf3.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-01-14 10:30:13 UTC; 2min 1s ago
       Docs: man:iperf3(1)
   Main PID: 4336 (iperf3)
      Tasks: 1 (limit: 4605)
     Memory: 612.0K (peak: 620.0K)
        CPU: 9ms
     CGroup: /system.slice/iperf3.service
             └─4336 /usr/bin/iperf3 --server --interval 0

Jan 14 10:29:58 group22-02 systemd[1]: iperf3.service: Main process exited, code=exited, status=217/USER
Jan 14 10:29:58 group22-02 systemd[1]: iperf3.service: Failed with result 'exit-code'.
Jan 14 10:30:13 group22-02 systemd[1]: iperf3.service: Scheduled restart job, restart counter is at 1.
Jan 14 10:30:13 group22-02 systemd[1]: Started iperf3.service - iperf3 server.
```

6.2 On Ubuntu Desktop install:

6.2.1 speedtest-cli

The speedtest-cli tool is installed to measure Internet throughput between the desktop system and an external speedtest server.

```
group22@group22-01:~$ sudo apt install speedtest-cli -y
[sudo] password for group22:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  speedtest-cli
0 upgraded, 1 newly installed, 0 to remove and 148 not upgraded.
Need to get 24.1 kB of archives.
After this operation, 106 kB of additional disk space will be used.
Get:1 http://al.archive.ubuntu.com/ubuntu noble/universe amd64 speedtest-cli all 2.1.3-2 [24.1 kB]
Fetched 24.1 kB in 0s (65.9 kB/s)
Selecting previously unselected package speedtest-cli.
(Reading database ... 150559 files and directories currently installed.)
Preparing to unpack .../speedtest-cli_2.1.3-2_all.deb ...
Unpacking speedtest-cli (2.1.3-2) ...
Setting up speedtest-cli (2.1.3-2) ...
Processing triggers for man-db (2.12.0-4build2) ...
```

6.2.2 curl

The curl utility is installed, allowing HTTP requests to be sent from the command line for testing web services.

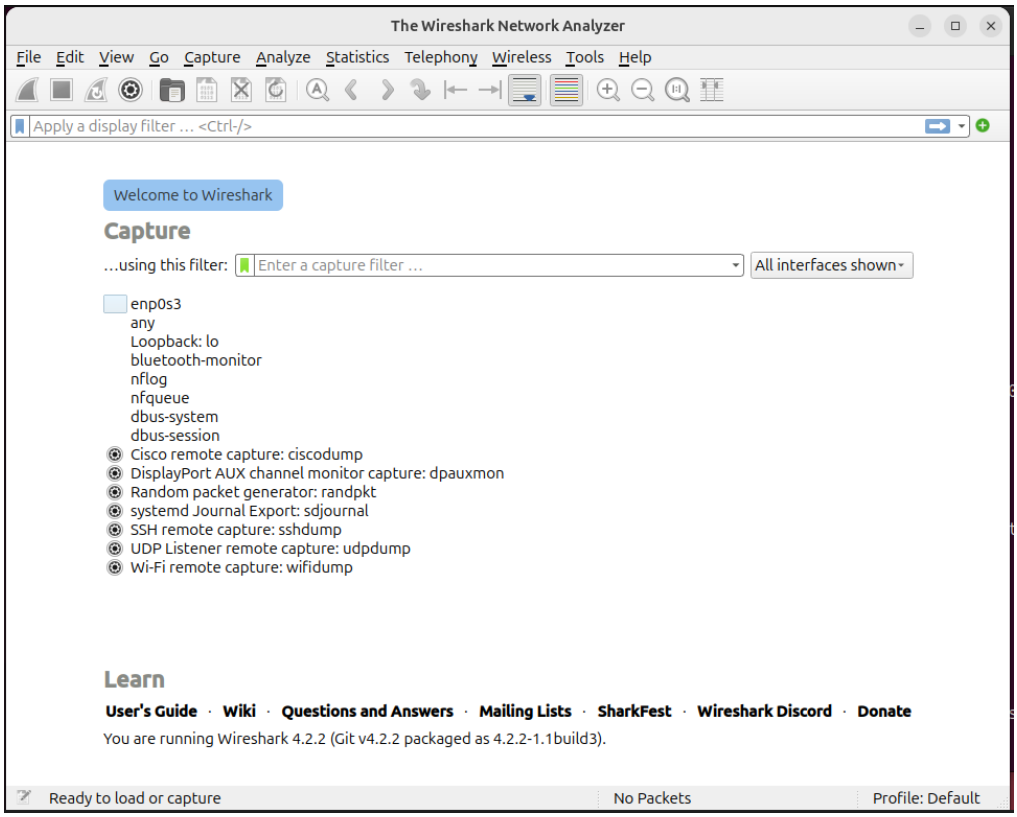
```
group22@group22-01:~$ sudo apt install curl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  curl
0 upgraded, 1 newly installed, 0 to remove and 148 not upgraded.
Need to get 226 kB of archives.
After this operation, 534 kB of additional disk space will be used.
Get:1 http://al.archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0-2ubuntu10.6 [226 kB]
Fetched 226 kB in 0s (602 kB/s)
Selecting previously unselected package curl.
(Reading database ... 150574 files and directories currently installed.)
Preparing to unpack .../curl_8.5.0-2ubuntu10.6_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu10.6) ...
Setting up curl (8.5.0-2ubuntu10.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
```

6.2.3 Wireshark

Wireshark is installed with all required dependencies, enabling packet capture and protocol analysis.

```
group22@group22-01:~$ sudo apt install wireshark -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  i965-va-driver intel-media-va-driver libaacs0 libavcodec60 libavformat60 libavutil58 libb2-1 libbcg729-0
  libbdplus0 libbluray2 libchromaprint1 libcjson1 libcodec2-1.2 libdav1d7 libdouble-conversion3 libegl-mesa0
  libgbm1 libgl1-mesa-dri libglx-mesa0 libgme0 libgsm1 libhwy1t64 libigdgmm12 libjxl0.7 libllvm20 liblua5.2-0
  libmbedcrypto7t64 libmd4c0 libminizip1t64 libnghttp3-3 libnorm1t64 libopencore-amrnb0 libopenmpt0t64
  libpcre2-16-0 libpgm-5.3-0t64 libqt6core5compat6 libqt6core6t64 libqt6dbus6t64 libqt6gui6t64
  libqt6multimedia6 libqt6network6t64 libqt6opengl6t64 libqt6printsupport6t64 libqt6qml6 libqt6qmlmodels6
  libqt6quick6 libqt6svg6 libqt6waylandclient6 libqt6waylandcompositor6 libqt6waylandeglclienthwintegration6
  libqt6waylandeglcompositorhwintegration6 libqt6widgets6t64 libqt6wlshellintegration6 librabbitmq4 librav1e0
  librist4 libshine3 libsmi2t64 libsnappy1v5 libsodium23 libsoxr0 libspandsp2t64 libsrt1.5-gnutls
  libssh-gcrypt-4 libsvtav1enc1d1 libswresample4 libswscale7 libts0t64 libudfread0 libva-drm2 libva-x11-2
  libva2 libvdpau1 libvpl2 libwireshark-data libwireshark17t64 libwiretap14t64 libwsutil15t64 libx264-164
  libx265-199 libxatracker2 libxvidcore4 libzmq5 libzvbi-common libzvbi0t64 mesa-libgallium mesa-va-drivers
  mesa-vdpau-drivers mesa-vulkan-drivers ocl-icd-libopencl1 qt6-gtk-platformtheme qt6-qpa-plugins
  qt6-translations-l10n qt6-wayland va-driver-all vdpau-driver-all wireshark-common
Suggested packages:
  i965-va-driver-shaders libcuda1 libnvcuvid1 libnvidia-encode1 libbluray-bdj qt6-qmltooling-plugins
  snmp-mibs-downloader geoipupdate geoip-database geoip-database-extra libjs-leaflet
  libjs-leaflet.markercluster wireshark-doc opencl-icd libvdpau-va-gl1
The following NEW packages will be installed:
  i965-va-driver intel-media-va-driver libaacs0 libavcodec60 libavformat60 libavutil58 libb2-1 libbcg729-0
  libbdplus0 libbluray2 libchromaprint1 libcjson1 libcodec2-1.2 libdav1d7 libdouble-conversion3 libgme0 libgsm1
  libhwy1t64 libigdgmm12 libjxl0.7 libllvm20 liblua5.2-0 libmbedcrypto7t64 libmd4c0 libminizip1t64 libnghttp3-3
```

Wireshark is launched successfully, and the active network interface **enp0s3** is available for packet capture.



7

7. Measure Performance and Document Results

7.1 Measure throughput

7.1.1 Between Ubuntu Desktop and Speedtest.net server using speedtest-cli or web browser

Speedtest-cli measures Internet performance, reporting **51.48 Mbit/s download**, **18.99 Mbit/s upload**, and **49.614 ms ping** to the selected test server. These values represent end-to-end throughput through the NAT Network and host Internet connection.

```
group22@group22-02:~$ speedtest-cli
Retrieving speedtest.net configuration...
Testing from One Albania (185.156.14.166)...
Retrieving speedtest.net server list...
Selecting best server based on ping...
Hosted by A1 (Sofia) [328.22 km]: 49.614 ms
Testing download speed.....................
Download: 51.48 Mbit/s
Testing upload speed.......................
..
Upload: 18.99 Mbit/s
```

7.1.2 Between Ubuntu Desktop and Ubuntu Server using iperf3 or netperf

From the desktop, iperf3 measures throughput to the server at **222.22.22.10**, achieving approximately **3.03 Gbits/sec**, indicating high-speed local network performance.

```
group22@group22-01:~$ iperf3 -c 222.22.22.10
Connecting to host 222.22.22.10, port 5201
[  5] local 222.22.22.11 port 34002 connected to 222.22.22.10 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec   449 MBytes  3.76 Gbits/sec  450    410 KBytes
[  5]   1.00-2.00   sec   468 MBytes  3.92 Gbits/sec  360    341 KBytes
[  5]   2.00-3.00   sec   435 MBytes  3.65 Gbits/sec  450    419 KBytes
[  5]   3.00-4.00   sec   491 MBytes  4.12 Gbits/sec  225    491 KBytes
[  5]   4.00-5.00   sec   301 MBytes  2.52 Gbits/sec  450    296 KBytes
[  5]   5.00-6.00   sec   294 MBytes  2.46 Gbits/sec  360    255 KBytes
[  5]   6.00-7.00   sec   308 MBytes  2.59 Gbits/sec  316    332 KBytes
[  5]   7.00-8.00   sec   296 MBytes  2.48 Gbits/sec  381    206 KBytes
[  5]   8.00-9.00   sec   282 MBytes  2.36 Gbits/sec  341    269 KBytes
[  5]   9.00-10.00  sec   289 MBytes  2.43 Gbits/sec  353    288 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-10.00  sec  3.53 GBytes  3.03 Gbits/sec  3686             sender
[  5]   0.00-10.00  sec  3.52 GBytes  3.03 Gbits/sec                  receiver

iperf Done.
```

From the server, iperf3 measures throughput to the desktop at **222.22.22.11**, achieving approximately **2.65 Gbits/sec**, confirming consistent bidirectional LAN performance.

```
group22@group22-02:~$ iperf3 -c 222.22.22.11
Connecting to host 222.22.22.11, port 5201
[  5] local 222.22.22.10 port 51774 connected to 222.22.22.11 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec   469 MBytes  3.93 Gbits/sec  540    477 KBytes
[  5]   1.00-2.00   sec   489 MBytes  4.11 Gbits/sec  315    383 KBytes
[  5]   2.00-3.00   sec   443 MBytes  3.72 Gbits/sec  585    191 KBytes
[  5]   3.00-4.00   sec   449 MBytes  3.77 Gbits/sec  622    508 KBytes
[  5]   4.00-5.00   sec   492 MBytes  4.13 Gbits/sec  495    342 KBytes
[  5]   5.00-6.00   sec   445 MBytes  3.73 Gbits/sec  327    272 KBytes
[  5]   6.00-7.00   sec   292 MBytes  2.45 Gbits/sec  192    540 KBytes
[  5]   7.00-8.00   sec  31.8 MBytes   266 Mbits/sec   45    440 KBytes
[  5]   8.00-9.00   sec  25.0 MBytes   210 Mbits/sec    0    486 KBytes
[  5]   9.00-10.00  sec  29.9 MBytes   250 Mbits/sec    0    520 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-10.00  sec  3.09 GBytes  2.66 Gbits/sec  3121             sender
[  5]   0.00-10.02  sec  3.09 GBytes  2.65 Gbits/sec                  receiver

iperf Done.
```

7.2 Measure Round Trip Time (RTT):

7.2.1 Between Ubuntu Desktop and Ubuntu Server

Ping measurements show an average round-trip time of approximately **1.236 ms**, reflecting very low latency within the local virtual network.

```
group22@group22-01:~$ ping -c 5 222.22.22.10
PING 222.22.22.10 (222.22.22.10) 56(84) bytes of data.
64 bytes from 222.22.22.10: icmp_seq=1 ttl=64 time=1.29 ms
64 bytes from 222.22.22.10: icmp_seq=2 ttl=64 time=1.13 ms
64 bytes from 222.22.22.10: icmp_seq=3 ttl=64 time=1.09 ms
64 bytes from 222.22.22.10: icmp_seq=4 ttl=64 time=1.64 ms
64 bytes from 222.22.22.10: icmp_seq=5 ttl=64 time=1.04 ms

--- 222.22.22.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4008ms
rtt min/avg/max/mdev = 1.043/1.236/1.639/0.217 ms
```

7.2.2 Between both hosts and a chosen public host

Ping from the desktop to a public host reports an average RTT of approximately **23.554 ms**, representing Internet latency.

```
group22@group22-01:~$ ping -c 5 www.google.com
PING www.google.com (142.251.140.36) 56(84) bytes of data.
64 bytes from sof04s05-in-f4.1e100.net (142.251.140.36): icmp_seq=1 ttl=255 time=23.0 ms
64 bytes from sof04s05-in-f4.1e100.net (142.251.140.36): icmp_seq=2 ttl=255 time=23.7 ms
64 bytes from sof04s05-in-f4.1e100.net (142.251.140.36): icmp_seq=3 ttl=255 time=24.2 ms
64 bytes from sof04s05-in-f4.1e100.net (142.251.140.36): icmp_seq=4 ttl=255 time=24.1 ms
64 bytes from sof04s05-in-f4.1e100.net (142.251.140.36): icmp_seq=5 ttl=255 time=22.7 ms

--- www.google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 22.730/23.554/24.240/0.583 ms
```

Ping from the server reports an average RTT of approximately **15.090 ms**, demonstrating normal variation due to routing and DNS resolution.

```
group22@group22-02:~$ ping -c 5 www.google.com
PING www.google.com (142.251.38.228) 56(84) bytes of data.
64 bytes from lcsofb-ab-in-f4.1e100.net (142.251.38.228): icmp_seq=1 ttl=255 time=15.0 ms
64 bytes from lcsofb-ab-in-f4.1e100.net (142.251.38.228): icmp_seq=2 ttl=255 time=15.4 ms
64 bytes from lcsofb-ab-in-f4.1e100.net (142.251.38.228): icmp_seq=3 ttl=255 time=15.1 ms
64 bytes from lcsofb-ab-in-f4.1e100.net (142.251.38.228): icmp_seq=4 ttl=255 time=15.4 ms
64 bytes from lcsofb-ab-in-f4.1e100.net (142.251.38.228): icmp_seq=5 ttl=255 time=14.5 ms

--- www.google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 6535ms
rtt min/avg/max/mdev = 14.531/15.090/15.430/0.326 ms
```

7.3 Use Wireshark to capture:

7.3.1 HTTP request/response packets between Ubuntu Desktop and Server

Wireshark captures HTTP GET requests from **222.22.22.11** to **222.22.22.10** and corresponding **HTTP/1.1 200 OK** responses, confirming successful application-layer communication between client and server.

http

No.	Time	Source	Destination	Protocol	Length	Info
155	1.282595290	222.22.22.11	34.107.221.82	HTTP	355	GET /canonical.html HTTP/1.1
157	1.305216088	34.107.221.82	222.22.22.11	HTTP	352	HTTP/1.1 200 OK (text/html)
243	1.959968836	222.22.22.11	216.58.212.35	OCSP	496	Request
245	2.079420795	216.58.212.35	222.22.22.11	OCSP	1157	Response
935	4.314983670	222.22.22.11	216.58.212.35	OCSP	490	Request
1217	4.397165841	216.58.212.35	222.22.22.11	OCSP	965	Response
1450	4.797170054	222.22.22.11	34.107.221.82	HTTP	372	GET /success.txt?ipv4 HTTP/1.1
1452	4.821083437	34.107.221.82	222.22.22.11	HTTP	270	HTTP/1.1 200 OK (text/plain)
1648	23.341448462	222.22.22.11	222.22.22.10	HTTP	406	GET / HTTP/1.1
1650	23.349307661	222.22.22.10	222.22.22.11	HTTP	733	HTTP/1.1 200 OK (text/html)
1652	23.386500639	222.22.22.11	222.22.22.10	HTTP	421	GET /favicon.ico HTTP/1.1
1653	23.388037948	222.22.22.10	222.22.22.11	HTTP	395	HTTP/1.1 404 Not Found (text/html)

7.3.2 DNS packets between Ubuntu Desktop and your chosen public website

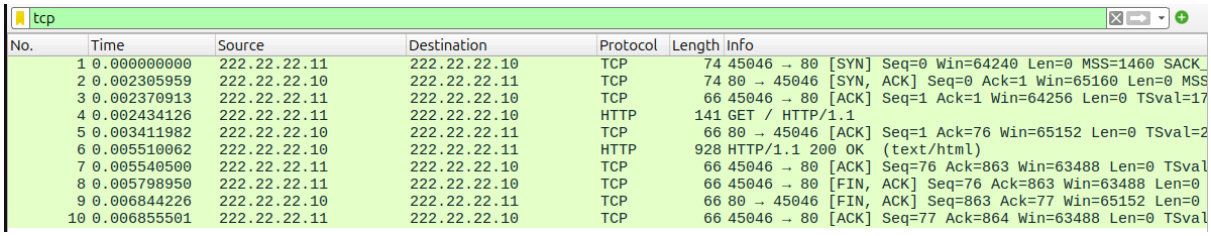
DNS packets show queries from **222.22.22.11** to **8.8.8.8** for **www.google.com**, followed by valid DNS responses, confirming correct DNS operation.

dns

No.	Time	Source	Destination	Protocol	Length	Info
111	13.718745615	222.22.22.11	8.8.8.8	DNS	85	Standard query 0xa405 A www.google.com OPT
112	13.740584755	8.8.8.8	222.22.22.11	DNS	101	Standard query response 0xa405 A www.google.com A 142
115	13.756866460	222.22.22.11	8.8.8.8	DNS	98	Standard query 0xc3eb PTR 228.38.251.142.in-addr.arpa
116	13.784278939	8.8.8.8	222.22.22.11	DNS	137	Standard query response 0xc3eb PTR 228.38.251.142.in-

7.3.3 A TCP three-way handshake between Ubuntu Desktop and Server

Wireshark captures the TCP three-way handshake consisting of **SYN**, **SYN-ACK**, and **ACK** packets between the desktop and server on TCP port **80**, demonstrating proper connection establishment before HTTP data transfer.



tcp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	222.22.22.11	222.22.22.10	TCP	74	45046 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_
2	0.002305959	222.22.22.10	222.22.22.11	TCP	74	80 → 45046 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS
3	0.002370913	222.22.22.11	222.22.22.10	TCP	66	45046 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=17
4	0.002434126	222.22.22.11	222.22.22.10	HTTP	141	GET / HTTP/1.1
5	0.003411982	222.22.22.10	222.22.22.11	TCP	66	80 → 45046 [ACK] Seq=1 Ack=76 Win=65152 Len=0 TSval=2
6	0.005510062	222.22.22.10	222.22.22.11	HTTP	928	HTTP/1.1 200 OK (text/html)
7	0.005540500	222.22.22.11	222.22.22.10	TCP	66	45046 → 80 [ACK] Seq=76 Ack=863 Win=63488 Len=0 TSval
8	0.005798950	222.22.22.11	222.22.22.10	TCP	66	45046 → 80 [FIN, ACK] Seq=76 Ack=863 Win=63488 Len=0
9	0.006844226	222.22.22.10	222.22.22.11	TCP	66	80 → 45046 [FIN, ACK] Seq=863 Ack=77 Win=65152 Len=0
10	0.006855501	222.22.22.11	222.22.22.10	TCP	66	45046 → 80 [ACK] Seq=77 Ack=864 Win=63488 Len=0 TSval

7.4 Use curl on Ubuntu Desktop to send an HTTP GET request to the Nginx/Apache server on Ubuntu Server

A curl HTTP GET request to **http://222.22.22.10** successfully retrieves the default Nginx HTML page. Curl acts as an HTTP client, and Nginx responds as the HTTP server over TCP port 80.

```
group22@group22-01:~$ curl http://222.22.22.10
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

8

8. Document the Project

8.1 For each task, explain concepts and terms from Computer Networks

Application Layer – User and Service Interaction

The Application Layer is responsible for providing network services directly to users and applications.

- **Nginx / Apache Web Server**
  Nginx (and alternatively Apache) operates at the Application Layer as a web server. It handles HTTP requests sent by the client and returns HTTP responses containing web content. In this project, Nginx functions as the server-side application, while the client initiates communication using HTTP.
- **Google Chrome / Mozilla Firefox**
  Web browsers act as HTTP clients. They generate HTTP GET requests, send them to the web server, and interpret the received HTTP responses for display.
- **OpenSSH Server and OpenSSH Client**
  The OpenSSH server provides secure remote access services, while the OpenSSH client initiates encrypted connections. Together, they enable secure command-line communication between client and server using the SSH protocol.
- **DNS and nslookup**
  DNS operates at the Application Layer by resolving domain names into IP addresses. The nslookup tool is used to query DNS servers and verify correct name resolution.
- **curl**
  Curl is a command-line HTTP client used to send HTTP requests directly to the web server, demonstrating application-layer communication without a graphical browser.

Transport Layer – Reliable and Secure Data Delivery

The Transport Layer ensures reliable communication between applications running on different hosts.

- **TCP (Transmission Control Protocol)**
  TCP is used by HTTP and SSH to provide reliable, ordered, and error-checked data transmission. Before data transfer begins, TCP establishes a connection using the three-way handshake.
- **TCP Three-Way Handshake**
  The handshake consists of SYN, SYN-ACK, and ACK packets. This process synchronizes sequence numbers and confirms that both endpoints are ready to communicate.
- **iperf3**
  Iperf3 measures Transport Layer performance by generating TCP traffic between client and server to evaluate throughput and retransmissions.

Network Layer – Logical Addressing and Routing

The Network Layer is responsible for logical addressing and packet forwarding between networks.

- **IP Addressing**
  Each device is assigned an IP address to uniquely identify it within the network. Both dynamic (DHCP) and static IP addressing are used to demonstrate flexible and fixed network configurations.
- **NAT Network**
  Network Address Translation allows multiple virtual machines with private IP addresses to access external networks using a single public address. The NAT Network acts as a virtual router and gateway.
- **Router (Default Gateway)**
  The router forwards packets between the local subnet and external networks. It determines the best path for packet delivery outside the local network.
- **Traceroute**
  Traceroute reveals the path packets take through the network by manipulating the IP Time-To-Live (TTL) field, helping analyze routing behavior.

Link Layer – Local Network Communication

The Link Layer handles communication between devices on the same physical or virtual network.

- **Switch**
  The switch forwards Ethernet frames based on MAC addresses, enabling communication between hosts within the same local network.
- **Ethernet Interfaces**
  Virtual network interfaces emulate physical network cards and allow devices to send and receive frames over the local network.

Cross-Layer Tools and Performance Metrics

Some tools and concepts provide insight across multiple layers rather than belonging to a single one.

- **Ping and RTT (Round Trip Time)**
  Ping uses ICMP to test reachability and measure RTT, reflecting network responsiveness and latency.
- **Wireshark**
  Wireshark captures and analyzes packets from multiple layers simultaneously, including DNS queries, HTTP traffic, and TCP handshakes. It provides visibility into how protocols interact across the stack.
- **DHCP (Dynamic Host Configuration Protocol)**
  DHCP automates IP configuration, bridging the Application and Network Layers by dynamically providing IP addresses, gateways, and DNS information.

8.2 Include screenshots to document all configurations and tests

Done ✔

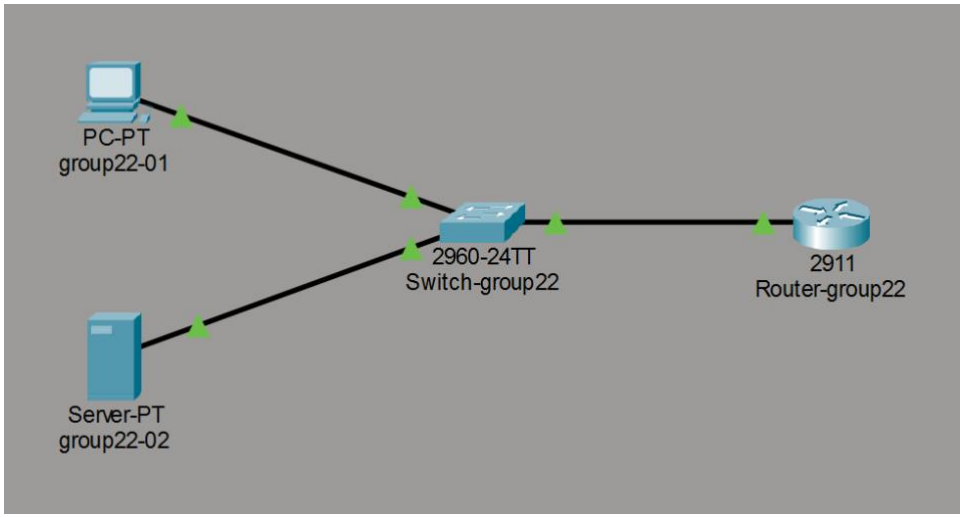
8.3 Use Cisco Packet Tracer to illustrate your network topology, including router, switch, and end-host configurations

Cisco Packet Tracer was used to illustrate the network topology implemented in this project. The simulated topology represents a simple local area network composed of a client, a server, a switch, and a router.

The topology consists of three main components:

- **End Hosts**
  - **PC-PT (group22-01):** Acts as the client machine. It represents a user workstation that initiates network communication, such as sending requests to the server.
  - **Server-PT (group22-02):** Acts as the server machine. It represents a service provider within the network, responding to client requests (e.g., web or SSH services).
- **Switch**
  - **2960-24TT (Switch-group22):** Functions as a Layer 2 device that interconnects the end hosts within the same local network. It forwards Ethernet frames based on MAC addresses and enables direct communication between the PC and the server.
- **Router**
  - **2911 (Router-group22):** Acts as the default gateway for the local network. It connects the LAN to external networks and enables routing beyond the local subnet.

All devices are connected using Ethernet links, and the active link indicators confirm correct physical and logical connectivity. This topology clearly demonstrates the interaction between end hosts, switching, and routing components, and supports the overall network design presented in this project.



9

9. Final Conclusion

This project successfully demonstrated the design, configuration, and evaluation of a **virtual network implementing a client–server architecture** using **VirtualBox**. A dedicated **NAT Network** with the subnet **222.22.22.0/24** was configured to simulate a private **Local Area Network (LAN)**, enabling controlled communication between virtual hosts while preserving external connectivity through **Network Address Translation (NAT)**.

Two Linux-based systems were deployed with clearly defined roles: **Ubuntu Server (group22-02)** as the service provider and **Ubuntu Desktop (group22-01)** as the client. Network configuration included **dynamic IP addressing via DHCP**, **static IP assignment**, **DNS configuration**, and **local hostname mapping**, ensuring consistent host identification and reliable communication. Connectivity and routing were verified using **ping (ICMP)**, **traceroute**, and **nslookup**, confirming correct functionality at the **Network** and **Transport layers**.

Application-layer services were installed and tested to validate real client–server interaction. **Nginx Web Server** operated as the **HTTP server**, handling **HTTP request/response** exchanges initiated by the **web browser** and **curl** acting as HTTP clients. Secure remote administration was achieved using the **OpenSSH Server** and **OpenSSH Client**, providing encrypted communication over **TCP**.

Network performance and behavior were evaluated using standard measurement and analysis tools. **iperf3** was used to measure **local network throughput**, while **speedtest-cli** assessed **external Internet performance** through the NAT environment. **Round Trip Time (RTT)** measurements highlighted the low latency of local communication compared to higher and variable latency when communicating with public hosts. Additionally, **Wireshark** was used to capture and analyze network traffic, including **DNS resolution**, **HTTP traffic**, and the **TCP three-way handshake**, directly linking theoretical concepts to packet-level evidence.

Finally, **Cisco Packet Tracer** was used to visually illustrate the logical network topology, including the **router**, **switch**, and **end hosts**, reinforcing the overall network design and providing a clear conceptual representation of device roles and connectivity. Overall, this project effectively integrated theoretical concepts from **Computer Networks (CEN 307)** with hands-on implementation, demonstrating how layered network communication is designed, deployed, tested, and analyzed in a controlled virtual environment.